COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences

PROGRAM (S) TO BE EVALUATED Computer Science

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	CS3002
Course Title	Information Security
Credit Hours	3
Prerequisites by Course(s) and Topics	Computer Networks (CS3001), Operating Systems (CS2006)
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assessment with the weight. **Assessment Type** / **Weight** Assignments: 5 Quiz: 10 Mid-Term: 30 Project: 10 Final: 45
Course Coordinator	Dr. Rana Asif Rehman
URL (if any)	
Current Catalog Description	Introduction to Information security, The CIA Triad: Confidentiality Integrity and Availability, Information security Models, Security compliance laws and regulations, Governance frameworks, Risk analysis, Security architectures, Malware classification. types of malware. Cryptography, Database & web security, Network security, Security policies,
Textbook (or **Laboratory Manual** for Laboratory Courses)	Cryptography Network Security: Principals and Practice, William Stallings Principle of Information Security, Whitman, Mattord Computer Security: Principals and Practice, William Stallings Hands-on Labs for Security Education, by SEED labs
Reference Material	Computer Security Fundamentals (second edition): Chuck Easttom

Program Learning Outcomes (PLOs):	This course covers the following PLOs: **PLO#** / **PLO Name** / **PLO Description** PLO 2 — Knowledge for Solving Computing Problems — Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements PLO 3 — Problem Analysis — Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines PLO 9 — Ethics — Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
Course Learning Outcomes (CLOs):	At the end of the course students will be able to: (see table below)

PLO#	PLO Name	PLO Description
PLO 2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
PLO 3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
PLO 9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice

At the end of the course students will be able to:

CLO #	CLO description	BT domain / BT Level	PLO #
CLO 1	**Explain** key concepts of information security such as design principles, cryptography, risk management	C2 (Understanding)	PLO 2
CLO 2	**Discuss** legal, ethical, and professional issues in information security	C2 (Understanding)	PLO 2
CLO 3	**Analyze** real world scenarios, model them using security measures, and apply various security and risk management tools for achieving information security and privacy	C4 (Analyzing)	PLO 3
CLO 4	**Identify** appropriate techniques to tackle and solve problems of real life in the discipline of information security	C2 (Understanding)	PLO 2
CLO 5	**Describe** issues related to ethics in the field of information security	C2 (Understanding)	PLO 9

\

Topics Covered in the Course, with Number of Lectures on Each Topic

Timeline	Content Covered
Lecture 1	**Course Introduction** • Introducing syllabus, policies, and projects.

(assume 15-week instruction and one-hour lectures)	• An overview of basic information security principles (with practical examples): confidentiality, integrity, availability, authentication, authorization and non-repudiation. • Component of an Information system **Lecture 2** **Security Design Principles** Discussion and evaluation of following primitives: Least-privilege, fail-safe defaults, complete mediation, separation of privilege, economy of mechanism, open design **Lecture 3** **Cryptography** Introduction to Cryptography: Symmetric cipher model, Substitution techniques (Caesar cipher, Monoalphabetic cipher) **Lecture 4** **Cryptography-II** Substitution techniques (Vigenere cipher, One-time pad) Transposition techniques (Rail fence cipher, Row transposition cipher) **Lecture 5** **Cryptography-II** Block cipher structure and design principle, Feistel cipher structure, the data encryption standard, DES (encryption, key generation) **Lecture 6** **Cryptography-III** AES structure, transformation, key expansion mechanism, AES example and implementation Stream ciphers introduction **Lecture 7** **Cryptography-IV** Introduction to Public Key cryptography RSA: principles, RSA algorithm Diffie-hellman key exchange algorithm with example, Man-in-the-middle attack in diffie-hellman **Lecture 8** **Cryptography-V** Hash functions, applications, Hash properties (preimage resistant, second preimage resistant, collision resistant) Message authentication code, requirements & properties of MAC HMAC algorithm & structure **Lecture 9** **Cryptography-VI** Digital Signature, requirements & properties of DS Public key infrastructure (PKI), elements of PKI, X.509, Digital certificates **Lecture 10** **Revision**

	First Mid-term Exam **Lecture 11** — **Software Security** Malware, types of malware (virus, worms, trojan horse, adware, spyware, backdoor, ransomware, rootkits, bootkits), malware analysis & countermeasures **Lecture 12** — **Software Security-II** Control Hijacking: Integer overflow String format vulnerabilities & countermeasures **Lecture 13** — **Software Security-III** Control Hijacking: Buffer overflow Countermeasures **Lecture 14** — **Database Security** Basics SQL Injection Attack, techniques, types of attack Countermeasures, database access control **Lecture 15** — **Database Security-II** Database inference attacks & counter measures Database encryption methods **Lecture 16** — **Web Security** Background Cross Site Request Forgery (CSRF) Attack Countermeasures (STP, origin header, referrer header) **Lecture 17** — **Web Security-II** Cross Site Scripting (XSS) Attack Types of XSS (reflected, stored, DOM based) countermeasures (encoding, validation, input handling contexts, secure input handing) **Lecture 18** — **User Authentication** Types (password, biometric, symmetric/asymmetric) Kerberos (overview, key exchange protocol) **Lecture 19** — **Access Control** Access control policies Discretionary, Attribute and Role-based Access Control **Lecture 20** — **Revision** **Second Mid-term Exam**

<table>
<tr><td></td><td colspan="4">
<table>
<tr><td>Lecture 21</td><td>Network Security
Secure Socket Layer (SSL)
SSL certificate, architecture, handshake</td></tr>
<tr><td>Lecture 22</td><td>Network Security-II
IP security (IPSec)
IPsec modes (transport, tunnel), architecture, AH, ESP</td></tr>
<tr><td>Lecture 23</td><td>Network Security-III
Intrusion Detection Systems (IDS)
Components of IDS, classification of IDS (anomaly, signature, hybrid), types (host-based, network based)</td></tr>
<tr><td>Lecture 24</td><td>Network Security-IV
Firewalls
Types of firewall (packet-filtering, stateful packet inspection, application proxy, circuit-level proxy)
Location of firewall</td></tr>
<tr><td>Lecture 25</td><td>Theoretical models of Access Control
Confidentiality policies (BLP model)
Integrity policies (Biba Model)
Integrity policies (Clark-Wilson model)
Hybrid policies (Chinese Wall model)</td></tr>
<tr><td>Lecture 26</td><td>Cybercrime Laws and Ethics
Pakistan cybercrime act and the role of investigative agencies.
Ethical perspective of research studies and experimentation (data privacy and anonymization techniques).
Intellectual property, copyright, patent, trade secret.</td></tr>
<tr><td>Lecture 27 - onwards</td><td>Revision & Project Evaluations</td></tr>
</table>
Final Examination
</td></tr>
<tr><td>Laboratory Projects/Experiments Done in the Course</td><td colspan="4"></td></tr>
<tr><td>Programming Assignments</td><td colspan="4">A programming assignment where students are expected to develop an application with a focus on identifying vulnerabilities and implementing mechanisms to address them.</td></tr>
<tr><td rowspan="2">Class Time Spent on
(in credit hours)</td><td>Theory</td><td>Problem Analysis</td><td>Solution Design</td><td>Social and Ethical Issues</td></tr>
<tr><td>40</td><td>25</td><td>25</td><td>10</td></tr>
</table>

Oral and Written Communications	Every student is required to submit at least __2___ written reports for the given assignments and to make __1___ oral presentations of typically __10___ minute's duration for the project. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.

Instructor Name: Dr. Ammar Haider

Instructor Signature